AP® STATISTICS SCORE-5 BLUEPRINT SERIES — VOLUME 1

AP® Statistics Formula & Inference Decision Guide

The High-Yield Score-5 Cram Companion: Rapid Formula Breakdowns, Inference Decision Trees, and Fill-in-the-Blank FRQ Sentence Frames

PR

2027 Exam Edition

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How to Use This Companion to Score a 5

Welcome to the **AP Statistics Formula & Inference Decision Guide**! This book was engineered specifically to solve the two biggest complaints students have with standard AP prep books:

1. **Textbook Bloat:** Standard books are 600+ pages of dense theory that take weeks to read.
2. **The FRQ Point Leak:** Students lose partial credit on Free-Response Questions not because their math is wrong, but because their verbal phrasing fails to satisfy strict AP rubric wording.

The 3-Step Score-5 System

Step 1: Master the Formulas & Parameters (Chapters 1–9): For every formula on the official College Board reference sheet, review the plain-English parameter translation, when to use it, and what conditions are required.

Step 2: Navigate the Master Inference Flowchart (Chapter 10): Before starting any hypothesis test or confidence interval problem, follow the decision tree to ensure you select the correct procedure without second-guessing.

Step 3: Eliminate Grader Traps & Memorize Sentence Frames (Chapters 11–12): Read through the Top 25 Exam Traps and adopt the exact fill-in-the-blank sentence frames for p-values, confidence levels, slope, and r^2 so that graders award you full points.

Pro Tip for the Final 14 Days Before Exam Day

Don't re-read entire textbooks. Spend 20 minutes each morning reviewing one unit's cheat sheet box and 10 minutes writing out the conditions checklist from memory.

Contents

Chapter 1

Unit 1: Exploring One-Variable Data

1.1 Essential Concepts & Cheat Sheet

When describing any quantitative single-variable distribution on the AP Exam, you **must** address four key characteristics: **SOCS** (Shape, Outliers, Center, Spread) always written **in context** of the problem.

Core Formulas: One-Variable Statistics

1. **Mean ($\bar{x}$):** $\bar{x} = \frac{\sum x_i}{n} = \frac{1}{n}\sum x_i$
2. **Sample Standard Deviation (s_x):** $s_x = \sqrt{\frac{1}{n-1}\sum(x_i - \bar{x})^2}$
3. **Standardized Score (z-score):** $z = \frac{x-\mu}{\sigma}$ or $z = \frac{x-\bar{x}}{s_x}$

1.2 Outlier Identification Rules

- **1.5 $\times$ IQR Rule:** $\text{IQR} = Q_3 - Q_1$. Outliers are values $< Q_1 - 1.5(\text{IQR})$ or $> Q_3 + 1.5(\text{IQR})$.
- **2-SD Rule (Normal data):** Outliers are values outside $\mu \pm 2\sigma$.

1.3 Resistant vs. Non-Resistant Summary Statistics

Measure	Resistant to Outliers / Skew?	When to Use on Exam
Median	**Yes** (Unaffected by extremes)	Skewed distributions
IQR	**Yes** (Middle 50% only)	Skewed distributions
Mean	**No** (Pulled toward tail/outliers)	Symmetric, unimodal distributions
Standard Deviation	**No** (Squared deviations inflate)	Symmetric, unimodal distributions

AP Exam Trap Alert: Comparing Distributions

When comparing two distributions on an FRQ:

1. You **must** use comparative words (e.g., *"Distribution A has a greater median than Distribution B"*).
2. You **must** include context (e.g., grams of sugar, test scores, heights).

Chapter 2

Unit 2: Exploring Two-Variable Data

2.1 Essential Concepts & Cheat Sheet

Core Formulas: Linear Regression

1. Least-Squares Regression Line (LSRL): $\hat{y} = a + bx$ ($\hat{y}$ = predicted response)
2. Slope (b): $b = r\left(\frac{s_y}{s_x}\right)$
3. y-Intercept (a): $a = \bar{y} - b\bar{x}$ (The point $(\bar{x}, \bar{y})$ is always on the line).
4. Residual (e): Residual = Actual y − Predicted $\hat{y} = y - \hat{y}$
5. Standard Error of Residuals (s): $s = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{n-2}}$

Score-5 Interpretation Sentence Templates

1. Slope (b): *"For each additional 1 [unit of x], the predicted [response variable y] increases/decreases by [value of b] [units of y]."*
2. r^2: *"Approximately [value of $r^2 \times 100\%$] of the variation in [response variable in context] is accounted for by the linear relationship with [explanatory variable in context]."*

Chapter 3

Unit 3: Collecting Data

3.1 Sampling Methods & Experimental Design

Sampling Method	Key Procedure & Purpose
SRS	Every group of n individuals has equal chance of selection. Eliminates selection bias.
Stratified	Homogeneous groups (*strata*); sample some from *all* strata. Reduces variability.
Cluster	Heterogeneous groups (*clusters*); randomly select several clusters and sample *all* within them. Saves cost.
Systematic	Select every k-th individual from an ordered list.

AP Exam Trap Alert: Confounding vs Bias

Confounding occurs when an outside variable is associated with the explanatory variable AND affects the response variable. Always explain both connections on FRQs!

Chapter 4

Unit 4: Probability, Random Variables, & Distributions

Core Probability Formulas

General Addition Rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional Probability: $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
Linear Combinations: $\mu_{X \pm Y} = \mu_X \pm \mu_Y, \quad \sigma^2_{X \pm Y} = \sigma^2_X + \sigma^2_Y$ (**Always ADD variances!**)
Binomial Distribution $B(n, p)$**:** $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \mu = np, \quad \sigma = \sqrt{np(1-p)}$

Chapter 5

Unit 5: Sampling Distributions

Sampling Distributions Formulas

Sample Proportion ($\hat{p}$): $\mu_{\hat{p}} = p, \quad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ (Normal if $np \geq 10, n(1-p) \geq 10$)
Sample Mean ($\bar{x}$): $\mu_{\bar{x}} = \mu, \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ (Normal if $n \geq 30$ by CLT or pop normal)

Chapter 6

Unit 6: Inference for Proportions

Proportions Inference Formulas

1-Prop z-Interval: $\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

1-Prop z-Test: $z = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$

2-Prop z-Test (Pooled): $\hat{p}_c = \frac{x_1+x_2}{n_1+n_2}, \quad z = \frac{(\hat{p}_1-\hat{p}_2)-0}{\sqrt{\hat{p}_c(1-\hat{p}_c)\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$

Chapter 7

Unit 7: Inference for Means

Means Inference Formulas

1-Sample t-Interval: $\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) \quad (\text{df} = n-1)$

1-Sample t-Test: $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}} \quad (\text{df} = n-1)$

2-Sample t-Test: $t = \frac{(\bar{x}_1-\bar{x}_2)-0}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}}$ (`Pooled = NO`)

Chapter 8

Unit 8: Chi-Square Tests

Chi-Square Formulas

$\chi^2 = \sum \frac{(O-E)^2}{E}$, Expected Count $= \frac{\text{(Row Total)}\times\text{(Column Total)}}{\text{Table Total}}$

Condition Alert

ALL Expected Counts MUST be ≥ 5 (never check observed counts).

Chapter 9

Unit 9: Inference for Slopes

Slope Inference

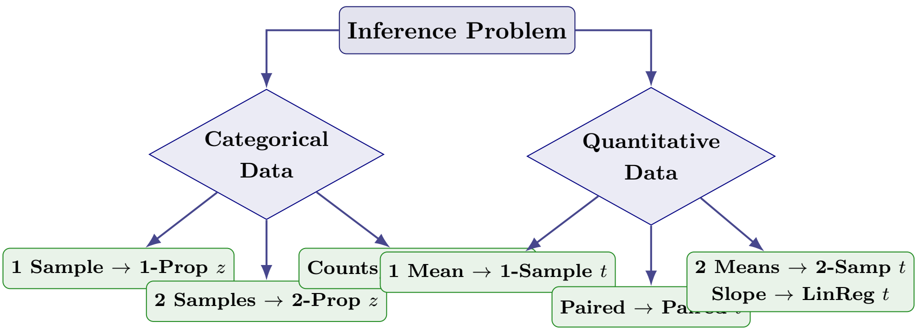
t-Interval for β: $b \pm t^* \cdot \mathrm{SE}_b \quad (\mathrm{df} = n - 2)$

t-Test for β: $t = \frac{b-0}{\mathrm{SE}_b} \quad (\mathrm{df} = n - 2)$

Chapter 10

Master Inference Decision Flowchart

10.1 Visual Inference Decision Tree



10.2 Inference Procedures Lookup Matrix

Data Type	**Groups**	**Goal**	**Procedure Name**	**Calculator Command**
Categorical	1 Sample	Estimate p	**1-Prop z-Interval**	`1-PropZInt`
Categorical	1 Sample	Test p	**1-Prop z-Test**	`1-PropZTest`
Categorical	2 Samples	Estimate $p_1 - p_2$	**2-Prop z-Interval**	`2-PropZInt`
Categorical	2 Samples	Test $p_1 = p_2$	**2-Prop z-Test (Pooled)**	`2-PropZTest`
Categorical	1 Sample (Fit)	Fit distribution	χ^2 **GOF-Test**	χ^2 `GOF-Test`
Categorical	2+ Samples	Homogeneity	χ^2 **Test of Homogeneity**	χ^2`-Test` (Matrix)
Categorical	1 Sample (2 Var)	Association	χ^2 **Test of Independence**	χ^2`-Test` (Matrix)
Quantitative	1 Sample	Estimate μ	**1-Sample t-Interval**	`TInterval`
Quantitative	1 Sample	Test μ	**1-Sample t-Test**	`T-Test`
Quantitative	Matched Pairs	Difference μ_d	**Paired t-Test/Interval**	`T-Test` on $L_1 - L_2$
Quantitative	2 Indep Samples	Compare $\mu_1 - \mu_2$	**2-Sample t-Test/Interval**	`2-SampTTest`
Quantitative	Linear Slope	Slope β	**LinReg t-Test/Interval**	`LinRegTTest`

Chapter 11

Top 25 "Never Forget" AP Exam Traps

1. **NEVER say "Accept H_0":** Always write *"Fail to reject H_0"*.
2. **NEVER write "We prove H_a":** Write *"We have convincing statistical evidence that [context]"*.
3. **Hypotheses in parameters (μ, p, β), NEVER statistics ($\bar{x}, \hat{p}, b$).**
4. **Define parameters in context.**
5. **Use comparative words when comparing distributions (greater, lesser).**
6. **Include units on ALL quantitative statistics and interpretations.**
7. **Don't forget the hat ($\hat{y}$) on regression equations.**
8. **Correlation (r) measures LINEAR strength only.**
9. **Association is NOT Causation.**
10. **Blocking is for experiments; Stratification is for sampling.**
11. **Mutually exclusive events are NEVER independent.**
12. **Always ADD variances, NEVER add standard deviations.**
13. **CLT applies to sampling distribution of $\bar{x}$ ($n \geq 30$), not individual data.**
14. **Use p_0 for 1-Prop Z-Test Large Counts ($np_0 \geq 10, n(1 - p_0) \geq 10$).**
15. **Use $\hat{p}$ for 1-Prop Z-Interval Large Counts ($n\hat{p} \geq 10, n(1 - \hat{p}) \geq 10$).**
16. **Chi-square conditions require EXPECTED counts ≥ 5, not observed counts.**
17. **Chi-square df:** $\text{df} = (\text{rows} - 1)(\text{cols} - 1)$.

18. **Never pool standard deviations for 2-sample t-tests (`Pooled = NO`).**
19. **Confidence Interval (plausible values) vs Confidence Level (long-run capture rate).**
20. **A high r^2 does NOT prove linearity (always check residual plot).**
21. **Voluntary response produces extreme bias.**
22. **State the conclusion in terms of H_a.**
23. **State the significance level (α).**
24. **Calculator Speak is NOT enough on FRQs (write formulas, df, p-value).**
25. **Context, Context, Context: Include problem context in every sentence!**

Chapter 12

Standard Interpretation Sentence Templates

Template 1: Confidence Interval for a Parameter

"We are [C%] confident that the interval from [Lower Bound] to [Upper Bound] captures the true [proportion / mean] of [context with units]."

Template 2: Confidence Level ($C\%$)

"If we take many random samples of size n and construct [C%] confidence intervals, about [C%] of the intervals will capture the true [parameter in context]."

Template 3: p-Value in Context

"Assuming that [null hypothesis H_0 in context] is true, there is a [p-value] probability of obtaining a sample statistic as extreme as or more extreme than observed, purely by random chance."

Template 4: Test Conclusion (Reject H_0)

"Because p-value ($[p]$) $<$ α ($[0.05]$), we reject H_0. There is convincing statistical evidence that [state H_a in context]."

Template 5: Test Conclusion (Fail to Reject H_0)

"Because p-value ($[p]$) $>$ α ($[0.05]$), we fail to reject H_0. There is NOT convincing statistical evidence that [state H_a in context]."